

Endpoint Security Incident Response and Automated Remediation Assessment Report

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Programme: MSc Cybersecurity
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Submission date: June 2026
Word count: Approximately 1,975 words excluding references and appendices
Repository: <https://github.com/syedhadihussain5125/CA1>

Academic Integrity Statement

I declare that this report and the associated lab artefacts are my own work. Third-party tools and reference material are acknowledged in the references. Generative AI was used to support wording and formatting only; the lab design, configuration, detection logic, testing, and analysis remain my own work.

Executive Summary

Context. Larkspur Retail Group is a fictional mid-sized retail organisation with Windows and Linux endpoints across office, warehouse, and remote staff environments. The scenario describes a breach involving credential theft, suspicious PowerShell activity, Linux privilege changes, and lateral movement risk.

Aim. The project set out to build an isolated endpoint security lab, forward Windows and Linux telemetry to a SIEM, detect simulated attacker activity, and demonstrate controlled automated remediation using a Dockerised AI security stack.

Approach. The lab uses three VMs on a VirtualBox host-only network: a Wazuh SIEM/AI host, one Windows 10 endpoint, and one Ubuntu 22.04 endpoint. Wazuh collects Security Event Log, Sysmon, PowerShell, auth.log, auditd, journald, and file integrity events. Six custom Wazuh detections are mapped to MITRE ATT&CK. Three active-response playbooks provide containment. A Docker Compose stack runs Ollama and Streamlit for read-only alert summarisation.

Key findings.

- Six custom detections cover brute force, Linux account creation, sudo escalation, Windows scheduled-task persistence, suspicious PowerShell, and sensitive privilege assignment.
- Three remediation actions are implemented: temporary IP blocking, Linux account locking, and Windows scheduled-task removal.
- The AI layer is intentionally advisory only. It cannot execute commands, access the Docker socket, or modify SIEM data.
- The lab design is safe because all attack simulations are benign and contained inside the host-only network.

Key deliverables.

- Wazuh all-in-one SIEM deployment and endpoint agent configurations.
- Windows and Linux telemetry configurations, including Sysmon and auditd.
- Custom Wazuh rules, simulation scripts, and active-response scripts.
- Dockerised AI alert dashboard using Ollama and Streamlit.
- Runbook, diagrams, evidence checklist, and video storyboard.

Impact and value. The project moves Larkspur from weak endpoint visibility to centralised detection and repeatable containment. It demonstrates a realistic open-source approach suitable for a small or medium organisation, while documenting the controls needed before production use.

Top recommendations.

Priority	Recommendation	Owner	Timeline
High	Re-enable Defender with targeted exclusions	IT Ops	Immediate
High	Replace weak lab passwords with managed credentials and MFA	IT Ops	Immediate
High	Enforce key-only SSH and remove password auth	Infrastructure	Immediate
Medium	Add automatic rollback approval for account locks	Security / IT Ops	30 days
Medium	Improve scheduled-task detection with command-path analysis	Security	45 days

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1 Introduction

1.1 Background

Endpoints are high-value targets because they store credentials, run user workloads, and provide paths to shared business systems. In the Larkspur Retail Group scenario, the attacker behaviours include credential theft, PowerShell execution, Linux privilege changes, and persistence. These behaviours require endpoint telemetry that is centralised, searchable, and connected to response procedures.

1.2 Objectives and Scope

The scope includes one Windows endpoint, one Linux endpoint, one Wazuh SIEM, and one Dockerised AI automation layer. The project covers lab isolation, telemetry forwarding, SIEM detections, benign attack simulations, automated remediation, validation, and rollback considerations.

1.3 Out of Scope

The project does not cover full enterprise segmentation, email gateway hardening, full disk forensics, domain-controller attack paths, real malware, or attacks against systems outside the lab.

1.4 Report Structure

The report moves from scenario and standards context to implementation, results, discussion, recommendations, and appendices containing the reproducibility artefacts.

2 Literature Review and Standards Alignment

Endpoint security relies on least privilege, logging, attack-surface reduction, and timely incident response. CIS Controls v8 emphasises audit log management and secure configuration. NIST SP 800-61 frames incident response as preparation, detection, containment, eradication, and recovery. MITRE ATT&CK provides a practical vocabulary for mapping detections to attacker techniques.

Sysmon improves Windows process visibility by capturing command line, parent process, hashes, and network metadata. auditd provides Linux syscall-level telemetry, including process execution and sensitive file changes. Wazuh provides a centralised agent-manager architecture, custom rule engine, MITRE mappings, and active-response hooks. Automated remediation is useful only when it is predefined, reversible, auditable, and proportionate to the alert severity. For that reason, the AI component in this project is limited to alert summarisation and classification support, while Wazuh active response performs deterministic remediation.

3 Scenario and Threat Model

3.1 Company Overview

Larkspur Retail Group is a retail organisation with approximately 200 endpoints supporting office users, warehouses, remote staff, shared file services, point-of-sale support systems, and customer order workflows.

3.2 Incident Narrative

The attacker first attempts SSH brute force against the Linux endpoint, then creates a backdoor local user and grants sudo privileges. On Windows, the attacker creates a suspicious scheduled task and also uses encoded PowerShell. These actions represent credential access, persistence, privilege escalation, and command execution.

3.3 Assumed Adversary

The assumed adversary is intermediate skill: able to use built-in operating-system tools, weak credentials, scheduled tasks, and PowerShell, but not assumed to use real malware or advanced evasion.

3.4 Assets, Trust Boundaries, and Data Classification

The key assets are endpoint credentials, endpoint process activity, authentication logs, and business-service access paths. The trust boundary is the isolated host-only network. Endpoint agents send encrypted telemetry to the Wazuh manager on TCP 1514. The Docker AI stack reads SIEM alerts in read-only mode.

3.5 Attack Paths and Hypotheses

Primary hypotheses are SSH brute force, local Linux account persistence, sudo escalation, Windows scheduled-task persistence, suspicious PowerShell execution, and privileged Windows logon activity.

4 Methodology

4.1 Lab Environment Build

Table 1: VM and IP plan

Host	OS	IP	Role	RAM
wazuh-manager	Ubuntu 22.04	192.168.56.10	Wazuh SIEM + AI stack	8 GB
windows-endpoint	Windows 10 Enterprise	192.168.56.20	Windows telemetry source	4 GB
linux-endpoint	Ubuntu 22.04	192.168.56.30	Linux telemetry source	4 GB

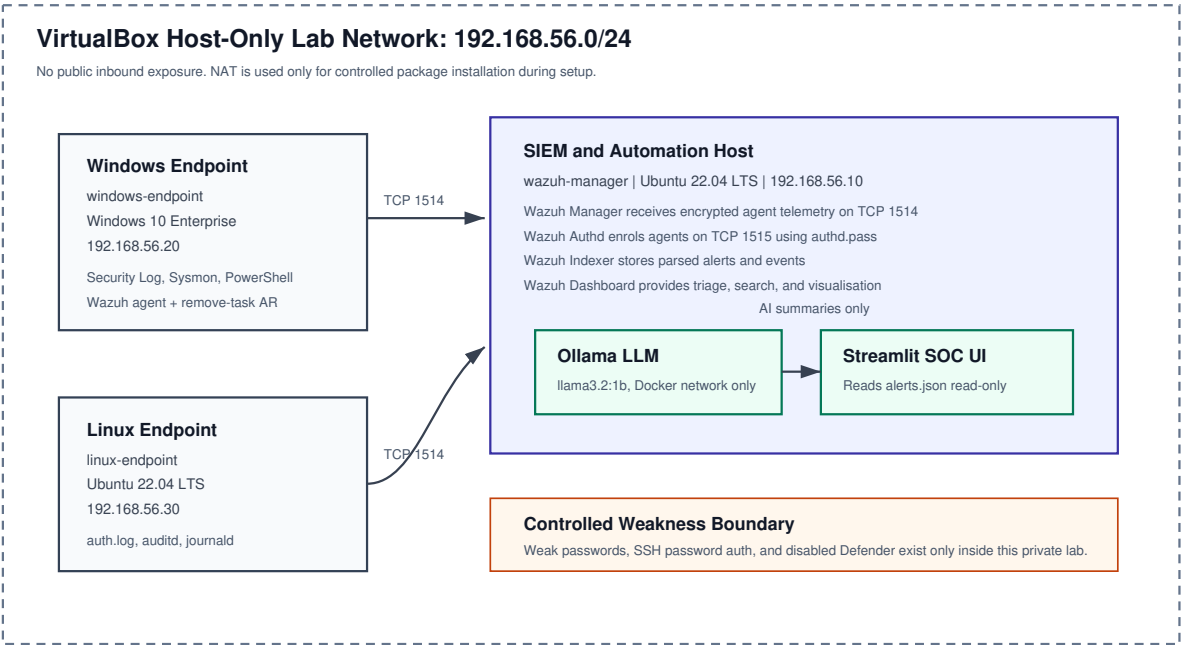


Figure 1: High-level lab architecture

4.2 Vulnerable Configuration Choices

Weaknesses are intentionally controlled: SSH password authentication is enabled, lab accounts use weak demo passwords, Windows Defender real-time monitoring is disabled to prevent simulation interference, and a Linux lab user has sudo capability. These weaknesses exist only in the private lab.

4.3 Telemetry Sources

Table 2: Telemetry sources

Platform	Source	Purpose
Windows	Security Event Log	Logon, process, account, privilege, and scheduled-task events
Windows	Sysmon	Process creation, network, file, registry, DNS, and tampering events
Windows	PowerShell Operational	Script block and module logging
Linux	auth.log	SSH, PAM, sudo, and user management events
Linux	auditd	execve, sudoers changes, sensitive file writes, cron changes
Linux	journal/syslog	System and service events

Telemetry Collection, Normalisation, and Triage Flow

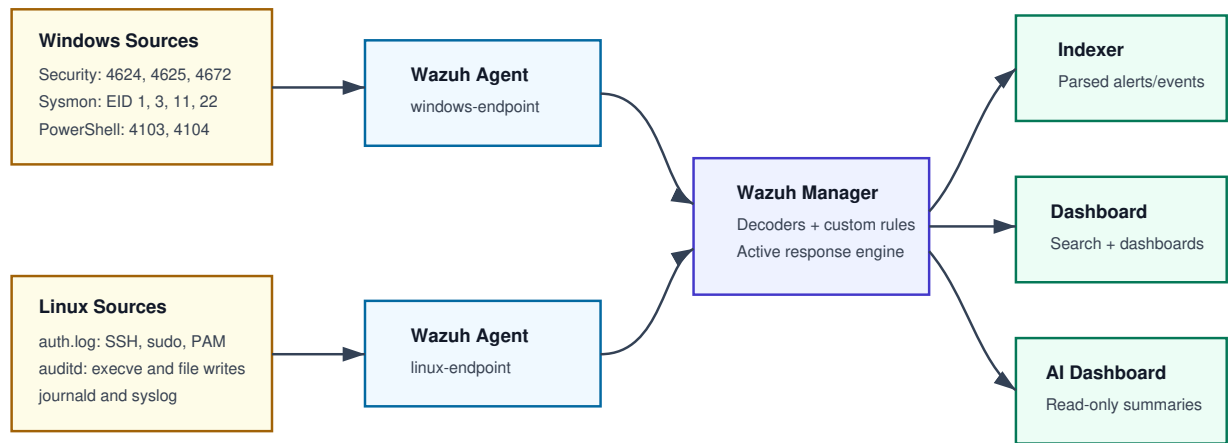


Figure 2: Telemetry and ingestion flow

4.4 SIEM Integration

Wazuh agents register to the manager using `authd.pass` and forward events over encrypted TCP 1514. Wazuh normalises fields such as `agent.name`, `rule.id`, `rule.level`, `data.srcip`, `data.dstuser`, `win.eventdata.commandLine`, and `timestamp`.

4.5 Detection Engineering Approach

Rules are mapped to MITRE ATT&CK and tuned to the lab scenario. Rule 100011 uses Wazuh correlation by chaining account creation to sudo escalation within a five-minute window, reducing false positives compared with single-event matching.

4.6 Automation and Remediation Design

Wazuh active response performs containment using allowlisted scripts. The AI dashboard has no write access to SIEM data, no Docker socket, no privileged mode, dropped Linux capabilities, and a read-only filesystem. It generates analyst summaries only.

Detection-to-Remediation Workflow

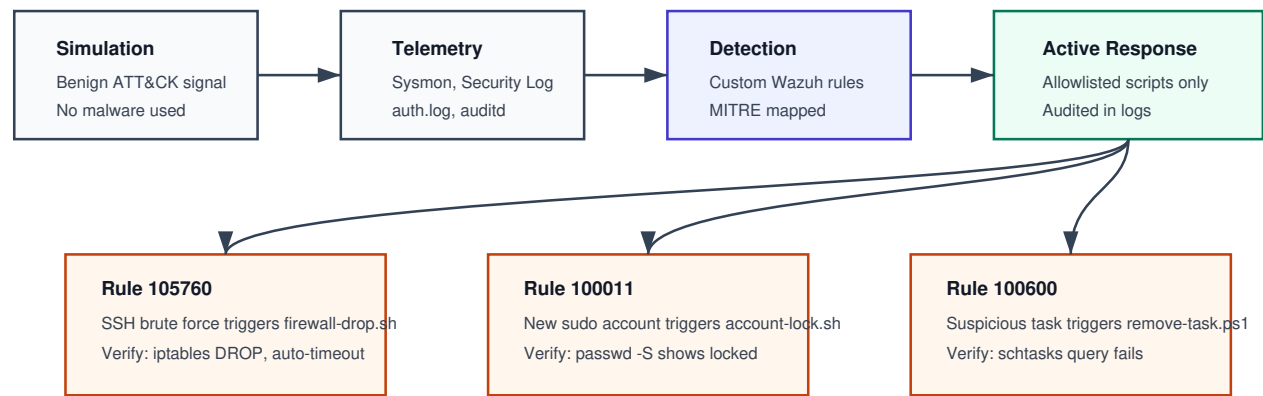


Figure 3: Detection-to-remediation workflow

4.7 Validation and Safety

Each test is run using benign scripts in `simulations/`. No real malware, third-party targets, or public attack infrastructure is used. Verification is performed with Wazuh alerts, active-response logs, and endpoint commands.

5 System Architecture and Implementation

The SIEM layer contains Wazuh Manager, Wazuh Indexer, and Wazuh Dashboard. The endpoint layer contains Wazuh agents, Sysmon/PowerShell logging on Windows, and auditd/auth.log/journald on Linux. The AI layer contains Ollama and a Streamlit dashboard connected by an internal Docker bridge network.

The Docker Compose configuration binds Ollama debug access to localhost only, mounts alerts read-only into the dashboard, uses non-root execution for the dashboard container, drops Linux capabilities, and applies memory limits. The active-response layer includes `firewall-drop.sh`, `account-lock.sh`, and `remove-task.cmd/ps1`.

6 Results

6.1 Detection Outcomes

Table 3: Detection outcomes

Rule	Use case	Source	ATT&CK	Result
105760	SSH brute force	auth.log	T1110	Alert + IP block
100010	New Linux user	auth.log/auditd	T1136.001	Alert
100011	Sudo escalation	auth.log/auditd	T1548.003	Alert + account lock
100600	Suspicious scheduled task	Windows EID 4698	T1053.005	Alert + task removal
100700	Suspicious PowerShell	Sysmon EID 1	T1059.001	Alert
100900	Sensitive privilege assignment	Windows EID 4672	T1078.003	Alert

6.2 Remediation Outcomes

Table 4: Remediation outcomes

Trigger	Action	Verification	Rollback
Rule 105760	firewall-drop.sh inserts iptables DROP	iptables -L INPUT -n --line-numbers	Auto-expires after 60 seconds
Rule 100011	account-lock.sh runs usermod -L	passwd -S backdooruser shows locked	usermod -U backdooruser
Rule 100600	remove-task.ps1 removes keylog task	schtasks /query /tn keylog fails	Recreate from exported XML

6.3 Performance, Reliability, and Limitations

Expected ingestion latency is suitable for a small lab. The design is limited by the flat lab network, disabled Defender, manual rollback for two actions, and the small Ollama model. Production deployment would require Wazuh clustering, certificate-based enrolment, stronger endpoint baselines, and more robust change approval.

7 Discussion

The detections are effective because they use strong telemetry sources. Sysmon provides command-line and parent-process context for PowerShell, while auditd provides Linux execution evidence. Correlation improves confidence for Linux privilege escalation because account creation followed quickly by sudo assignment is a stronger signal than either event alone.

The main trade-off is between coverage and noise. Broad telemetry can overwhelm analysts, so the lab forwards targeted event IDs while retaining the events needed for the simulated breach paths. Automation introduces operational risk if false positives trigger containment. The project mitigates this through allowlisted scripts, protected-account checks, short firewall timeouts, task export before deletion, and clear rollback instructions.

The design aligns with NIST SP 800-61 by supporting detection and containment, with CIS Controls through log management, and with MITRE ATT&CK through technique mapping. The AI stack improves triage readability but remains advisory to avoid unsafe autonomous command execution.

8 Recommendations

Table 5: Prioritised recommendations

Recommendation	Rationale	Priority	Owner	Timeline
Re-enable Defender with exclusions	Full Defender disablement is unsafe outside the lab	High	IT Ops	Immediate
Enforce key-only SSH and MFA	Weak password auth enables brute force	High	Infrastructure	Immediate
Add approval/rollback workflow	Account locks and task removals can disrupt users	High	Security / IT Ops	30 days
Tune detections with allowlists	Service accounts and admin scripts can create false positives	Medium	Security	45 days
Scale Wazuh and upgrade AI model	Production use needs HA, retention, and stronger summarisation	Medium	Security / Infra	60 days

9 Conclusion

The project delivers an isolated endpoint security lab with Windows and Linux telemetry, Wazuh SIEM detections, benign attack simulations, controlled automated remediation, and a Dockerised AI summarisation layer. The implementation satisfies the technical intent of the assignment and demonstrates an end-to-end workflow from signal generation to detection, triage, remediation, and verification.

The most important remaining submission task is evidence capture: the final PDF and video must show live Wazuh ingestion, detections firing, active-response verification, and AI summarisation from the running lab.

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Appendices

- Appendix A: Lab diagram: [report/figures/lab-architecture.svg](#)
- Appendix B: VM build and IP plan: [Table 1](#)
- Appendix C: Sysmon excerpt/hash reference: [wazuh/sysmon-config.xml](#)
- Appendix D: auditd rule set: [wazuh/auditd.rules](#)
- Appendix E: SIEM rules and queries: [wazuh/local_rules.xml](#) and Appendix F filters
- Appendix F: Required screenshot evidence checklist:

Evidence	Required Capture
Agent status	Wazuh Agents page showing both endpoints Active
Windows detections	agent.name:windows-endpoint AND (rule.id:100600 OR rule.id:100700 OR rule.id:100900)
Linux detections	agent.name:linux-endpoint AND (rule.id:105760 OR rule.id:100010 OR rule.id:100011)
Dashboard 1	Authentication anomalies view
Dashboard 2	Process execution anomalies view
Remediation log	tail -50 /var/ossec/logs/active-responses.log
IP block	sudo iptables -L INPUT -n --line-numbers
Account lock	sudo passwd -S backdooruser
Task removal	schtasks /query /tn keylog
AI summary	Streamlit dashboard summary for a selected high-severity alert

Appendix G: Docker Compose component descriptions: [docker-compose.yml](#)

Appendix H: Test cases and results are listed in [report/CA1-Report.md](#) Appendix H

Appendix I: Video storyboard is listed in [report/CA1-Report.md](#) Appendix I